

Pf lab post task 06

Q1: the rickshaw fare calculator:

```
#include <stdio.h>

int main()
{
    float distance, total_fare;
    int hour;

    printf("enter the distance\n");
    scanf("%f", &distance);
    printf("enter the hour (0-24)\n");
    scanf("%d", &hour);
    if (distance<=0)
    {
        printf("invalid distance\n");
        return 0;
    }
    total_fare = 50.0;
    if(distance>1.0)
    {
        total_fare += (distance - 1.0)*22.0;
    }
    if (hour<6 || hour>22)
```

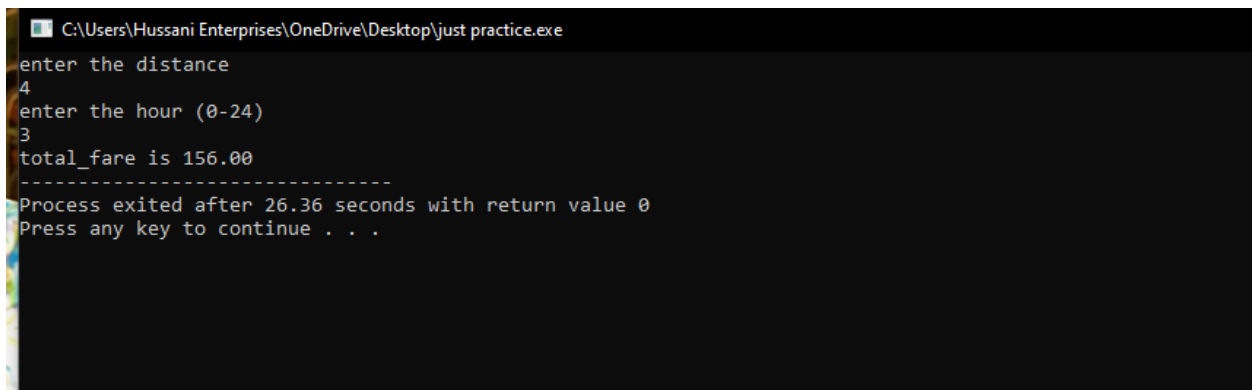
```

{
    total_fare += 40.0;
}

printf("total_fare is %.2f", total_fare);

return 0;

```



```

C:\Users\Hussani Enterprises\OneDrive\Desktop\just practice.exe
enter the distance
4
enter the hour (0-24)
3
total_fare is 156.00
-----
Process exited after 26.36 seconds with return value 0
Press any key to continue . . .

```

Q2: the grocery store discount slip

```

#include <stdio.h>

int main()
{
    float total_bill, discount_rate, discount_amount, final_payable;

    int membership_status;

    printf("enter the total-bill\n");

    scanf("%f", &total_bill);

    printf("do you have the membership_status 1.for yes 2.for no\n");

```

```
scanf("%d", &membership_status);
```

```

    if (total_bill < 500) {
        discount_rate = 0.0;
    } else
        if (total_bill >= 500 && total_bill <= 1999)
        {
            if (membership_status == 1)
            {
                discount_rate = 0.10;
            } else {
                discount_rate = 0.05;
            }
        } else if (total_bill >= 2000)
        {
            if (membership_status == 1)
            {
                discount_rate = 0.15;
            } else {
                discount_rate = 0.08;
            }
        }
    }
}

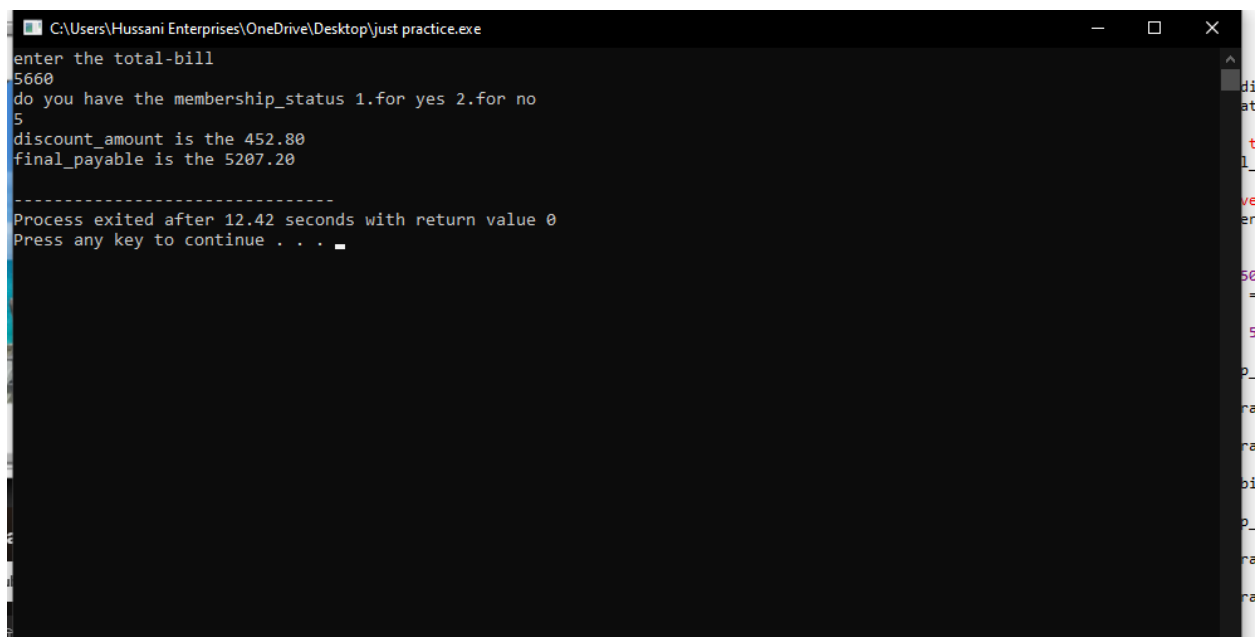
```

```
discount_amount= total_bill*discount_rate;
```

```
final_payable= total_bill-discount_amount;
```

```
printf("discounte_amount is the %.2f\n", discount_amount);
```

```
printf("final_payable is tthe %.2f\n", total_bill);
```



```
C:\Users\Hussani Enterprises\OneDrive\Desktop\just practice.exe
enter the total-bill
5660
do you have the membership_status 1.for yes 2.for no
5
discount_amount is the 452.80
final_payable is the 5207.20
-----
Process exited after 12.42 seconds with return value 0
Press any key to continue . . .
```

Q3: the exam geade slip

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
    int obtained_marks;
```

```
    printf("enter the obtained_marks\n");
```

```
scanf("%d", &obtained_marks);
```

```
if(obtained_marks > 100 || obtained_marks < 0)
```

```
{
```

```
    printf("invalid marks\n");
```

```
    return 0;
```

```
}
```

```
if(obtained_marks >= 90)
```

```
{
```

```
    printf("A+ grade\n");
```

```
    printf("Result:pass\n");
```

```
} else
```

```
if(obtained_marks >= 80 && obtained_marks <= 89)
```

```
{
```

```
    printf("A grade");
```

```
    printf("Result:pass\n");
```

```
} else
```

```
if(obtained_marks >= 70 && obtained_marks <= 79)
```

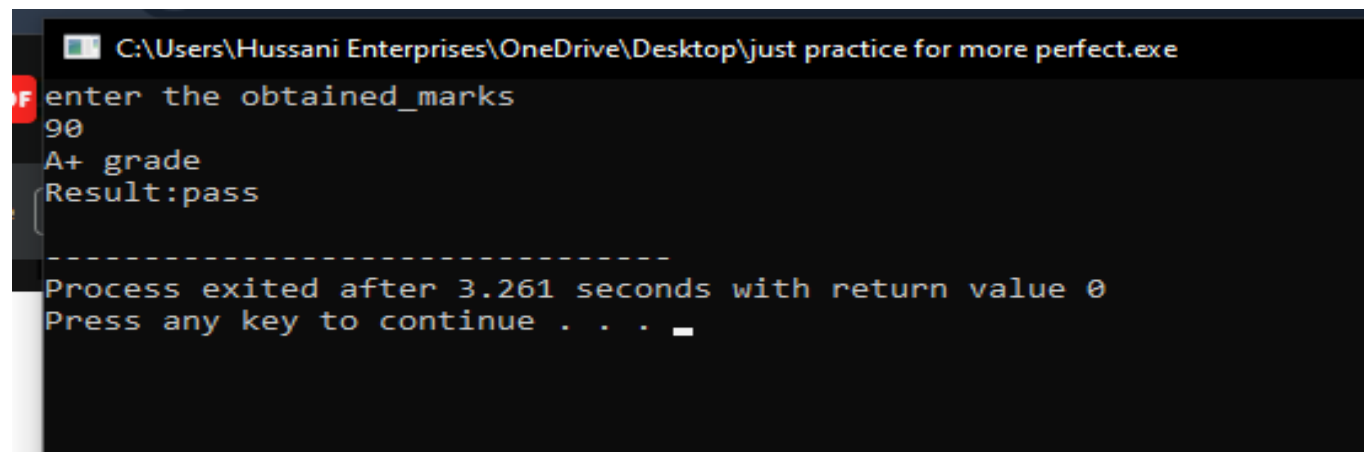
```
{
```

```
    printf("B grade\n");
```

```
    printf("Result:pass\n");
```

```
} else
```

```
if(obtained_marks>=60 && obtained_marks<=69)
{
    printf("C grade\n");
    printf("Result:pass\n");
} else
if(obtained_marks>=50 && obtained_marks<=59)
{
    printf("D grade\n");
    printf("Result:pass\n");
} else
{
    printf("fail\n");
    printf("Result: Fail\n");
}
}
```



```
C:\Users\Hussani Enterprises\OneDrive\Desktop\just practice for more perfect.exe
enter the obtained_marks
90
A+ grade
Result:pass
-----
Process exited after 3.261 seconds with return value 0
Press any key to continue . . .
```

Q4: The water tank timer:

```
#include <stdio.h>

int main()
{
    float tank_capacity,c_water_level, remaining_liters;
    float motorfill_rate, required_time,electricity_bill;

    printf("enter the tank_capacity in the liters:\n");
    scanf("%f", &tank_capacity);

    printf("enter the c_water_level in the liters:\n");
    scanf("%f", &c_water_level);

    printf("enter the motorfill_rate:\n");
    scanf("%f", &motorfill_rate);

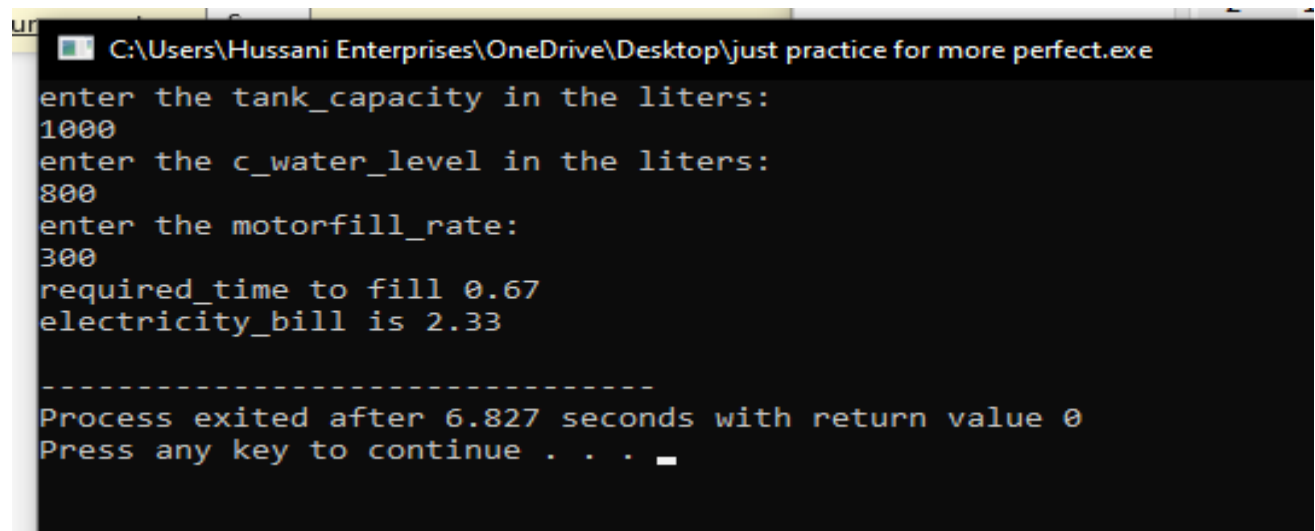
    if(tank_capacity == c_water_level)
    {
        printf("tank is already full\n");
        return 0;
    }

    remaining_liters = tank_capacity-c_water_level;
```

```

    required_time= remaining_liters/ motorfill_rate;
    electricity_bill= required_time*3.50;
    printf("required_time to fill %.2f\n", required_time);
    printf("electricity_bill is %.2f\n", electricity_bill);
}

```



```

C:\Users\Hussani Enterprises\OneDrive\Desktop\just practice for more perfect.exe
enter the tank_capacity in the liters:
1000
enter the c_water_level in the liters:
800
enter the motorfill_rate:
300
required_time to fill 0.67
electricity_bill is 2.33

-----
Process exited after 6.827 seconds with return value 0
Press any key to continue . . .

```

Q5: The mobile load card vendor;

```

#include <stdio.h>

int main(){
    float load_amount, bonus_percentage, bouns_amount, final_balance;
    int network_code, weekend_status;

    printf("enter the load_amount:\n");
    scanf("%f", &load_amount);

```

```
printf("enter the network_code : (1.jazz 2.telenor 3.ufone)\n");  
scanf("%d", &network_code);
```

```
printf("enter the weekend_status: 1.weekend 2.weekday\n");  
scanf("%d", &weekend_status);
```

```
    if (load_amount < 100.0) {  
        bonus_percentage = 0.0;  
    }  
    else {  
        if (load_amount <= 499.0) {  
            if (weekend_status == 1) {  
                if (network_code == 3) {  
                    bonus_percentage = 12.0;  
                }  
                else {  
                    bonus_percentage = 10.0;  
                }  
            }  
            else {  
                bonus_percentage = 5.0;  
            }  
        }  
    }
```

```
}  
else {  
    if (network_code == 1 || weekend_status == 1) {  
        bonus_percentage = 20.0;  
    }  
    else {  
        bonus_percentage = 15.0;  
    }  
}  
}  
  
bouns_amount= load_amount*(bonus_percentage/100);  
final_balance= load_amount + bouns_amount;  
  
printf("==receipt==\n");  
printf("load_amount is : %.2f\n", load_amount);  
    printf("bouns_percentage is : %.1f\n", bonus_percentage);  
        printf("bouns_amount is %.2f\n", bouns_amount);  
            printf("final_balance is %.2f", final_balance);  
}
```

```
C:\Users\Hussani Enterprises\OneDrive\Desktop\just practice for more perfect.exe
enter the load_amount:
550
enter the network_code : (1.jazz 2.telenor 3.ufone)
3
enter the weekend_status: 1.weekend 2.weekday
0
==receipt==
load_amount is : 550.00
bouns_percentage is : 15.0
bouns_amount is 82.50
final_balance is 632.50
-----
Process exited after 6.834 seconds with return value 0
Press any key to continue . . .
```

Q6: the library fine calculator:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int overdue_day, book_type, membership;
```

```
    float fine;
```

```
    printf("how many days late is it?\n");
```

```
    scanf("%d", &overdue_day);
```

```
    printf("1.regular\n2.reference\n3.rare\n");
```

```
    printf("enter the kind of the book\n");
```

```
    scanf("%d", &book_type);
```

```
printf("have you a membership?: 1.for yes 2.for no\n");
```

```
scanf("%d", &membership);
```

```
if(book_type == 1)
```

```
{
```

```
    if(overdue_day <=7)
```

```
    {
```

```
        fine= overdue_day*5.0;
```

```
    } else {
```

```
        fine = (7*5.0)+((overdue_day - 7)*10.0);
```

```
    }
```

```
} else if (book_type == 2)
```

```
{
```

```
    fine= overdue_day*15;
```

```
} else if (book_type == 3)
```

```
{
```

```
    fine = overdue_day* 30.0;
```

```
} if (membership == 1 && book_type != 3)
```

```
{
```

```
    fine = fine * 0.80;
```

```
}
```

```
printf("fine is %.2f", fine);
```

```

        if(overdue_day > 30)

        {

            printf("banned from borrowing\n");

        }

    }

```

```

C:\Users\Hussani Enterprises\OneDrive\Desktop\just practice for more perfect.exe
how many days late is it?
3
1.regular
2.reference
3.rare
enter the kind of the book
3
have you a membership?: 1.for yes 2.for no
1
fine is 90.00
-----
Process exited after 3.967 seconds with return value 0
Press any key to continue . . .

```

Q7: cricket team selector:

```

if (match_played < 5) {

    printf("Rejected — Insufficient Matches\n");

}

// RULE 2: Normal Selection Criteria

else if (batting_average >= 35 && match_played >= 10) {

    if (fitness_failure == 1) {

        printf("Rejected — Fitness\n");

    } else {

```

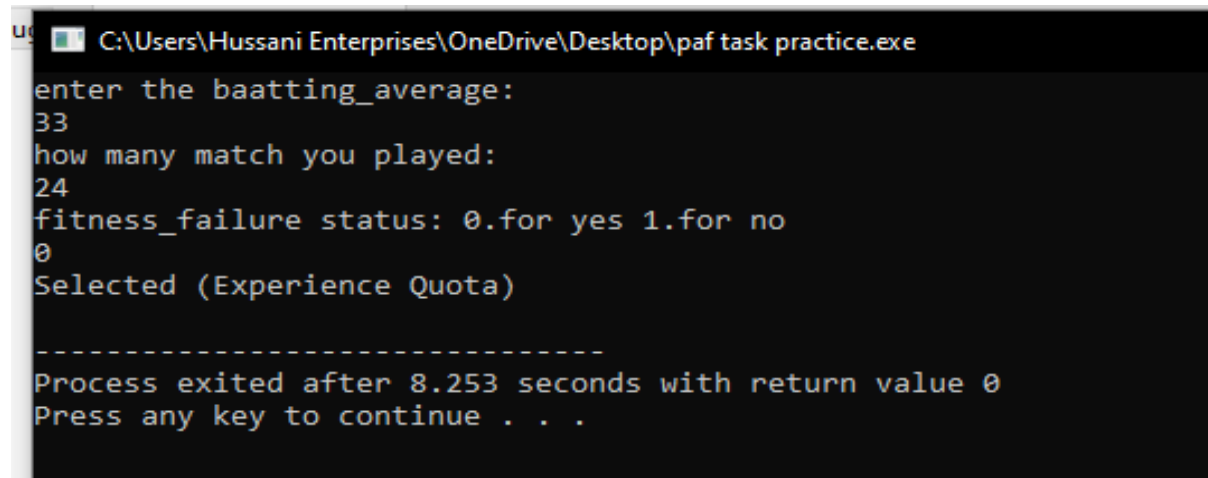
```

        printf("Selected\n");
    }
}

// RULE 3: Experience Quota Criteria
else if (batting_average >= 25 && batting_average <= 34.99 && match_played
>= 20) {
    if (fitness_failure == 1) {
        printf("Rejected — Fitness\n");
    } else {
        printf("Selected (Experience Quota)\n");
    }
}

// RULE 4: Default fallback if no conditions are met
else {
    printf("Not Selected\n");
}

```



```

C:\Users\Hussani Enterprises\OneDrive\Desktop\paf task practice.exe
enter the baatting_average:
33
how many match you played:
24
fitness_failure status: 0.for yes 1.for no
0
Selected (Experience Quota)

-----
Process exited after 8.253 seconds with return value 0
Press any key to continue . . .

```

Q8: the restaurant billing system:

```
#include <stdio.h>
```

```
int main ()
```

```
{
```

```
    int meal_category, bill_amount, customer_type;
```

```
    float service_charge, discount, final_bill;
```

```
    printf("enter the meal_category\n1.fast food\n2.desi food\n3.chines\n");
```

```
    scanf("%d", &meal_category);
```

```
    printf("enter the bill_amount\n");
```

```
    scanf("%d", &bill_amount);
```

```
    printf("enter the customer_type:\n1.student\n2.regular\n");
```

```
    scanf("%d", &customer_type);
```

```
    if( meal_category>3 || bill_amount<0 || customer_type>3)
```

```
    {
```

```
        printf("invalid info\n");
```

```
        return 0;
```

```
    }
```

```
    switch (meal_category)
```

```
{  
    case 1:  
        printf("==fast food==\n");  
        service_charge = 0.05 ;  
        break;  
    case 2:  
        printf("==desi food==\n");  
        service_charge = 0.08;  
        break;  
    case 3:  
        printf("==chines==\n");  
        service_charge = 0.1;  
        break;  
    default:  
        printf("invalid category\n");  
}
```

```
if(bill_amount>=1000)  
{  
    if(customer_type==1)  
    {  
        discount= bill_amount*0.15;  
    } else {
```

```
        discount= bill_amount*0.10;
    }
} else
if(bill_amount<1000)
{
    if(customer_type==1)
    {
        discount= bill_amount*0.05;
    } else
    {
        discount= 0.0;
    }
}

final_bill= bill_amount+service_charge-discount;

printf("\nbill_amount is %.2f", bill_amount);
printf("\ndiscount is %.2f", discount);
printf("\nfinal_bill is %.2f", final_bill);

return 0;

}
```

```
C:\Users\Hussani Enterprises\OneDrive\Desktop\Untitled1.exe
enter the meal_category
1.fast food
2.desi food
3.chines
3
enter the bill_amount
1100
enter the customer_type:
1.student
2.regular
1
==chines==

bill_amount is 0.00
discount is 165.00
final_bill is 935.10
-----
Process exited after 8.963 seconds with return value 0
Press any key to continue . . .
```